IIMB Management Review (2025) 37, 100570

available at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/iimb

Exploring predictive prowess of ensemble machine learning models in banking stocks: A technical, fundamental, and macroeconomic analysis

Sabyasachi Mohapatra[a], Rohan Mukherjee[b], Nicholas Apergis[c], Anirban Sengupta[a,*]

[a] *Indian Institute of Management, Bodhgaya, Bihar, India*
[b] *International Management Institute, Kolkata, West Bengal, India*
[c] *University of Derby, Derby, UK*

Received 13 February 2024; revised form 25 May 2024; accepted 29 April 2025; Available online 3 May 2025

KEYWORDS
Machine learning;
Stock prediction;
Technical indicators;
Firm-specific factors;
Macroeconomic variables

Abstract The work scrutinises the predictive prowess of ensemble machine learning models, namely Random Forest, Gradient Boosting, and XGBoost, in the domain of stock prediction by training models at two different stages. In stage 1, we restrict our evaluation to 18 technical indicators alongside holding period returns of 20 Indian Banks between January 2013 and March 2022, for short-term (20, 40, and 60 days), medium-term (180 days), and long-term durations (240 days); in stage 2, we further develop the study by including an additional combination of 6 firm-specific fundamental factors and 6 macroeconomic variables along with 18 technical indicators designed using price, volume, and momentum. During stage 1, we observe a modest range of performance, that is, between 62% and 78% across metrics, namely, accuracy score, F1 score, precision values, recall score, and specificity numbers. However, with the inclusion of fundamental and macroeconomic factors in stage 2, we observe a significant improvement in performance metrics to the tune of 90% across models for different holding periods. Particularly, for XGBoost, the reported accuracy range lies between 96% and 98%. Results indicate that while technical indicators are extremely important for short-term returns, the feature importance of fundamental and macroeconomic variables is highlighted during medium-term and long-term, respectively.
© 2025 Published by Elsevier Ltd on behalf of Indian Institute of Management Bangalore. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

*Corresponding author. Tel.: +91-8850109821.
E-mail address: anirbans@iimbg.ac.in (A. Sengupta).

https://doi.org/10.1016/j.iimb.2025.100570
0970-3896 © 2025 Published by Elsevier Ltd on behalf of Indian Institute of Management Bangalore. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

Introduction

Post the 2008 Global Financial Crisis, market participants questioned performance linked to traditional asset pricing approaches for stock selection. As traditional factor models cease to be less profitable, the focus entirely shifted to a 'successful prediction' based on dynamic learning from past data (Rasekhschaffe & Jones, 2019), which helps generate lucrative profits not limited to long-term investors, but also to short-term traders.

As discourses relating to efficient market hypothesis negate any scope for prediction considering information symmetry, the research community strived a lot to prove the discrepancy otherwise. In the process, traditional statistical methods, such as Autoregressive Conditional Heteroskedasticity (ARCH), Generalised Autoregressive Conditional Heteroskedasticity (GARCH), and Autoregressive Integrated Moving Average (ARIMA) gained appreciable popularity in the initial phase. However, statistical methods had their own limitations in forecasting returns, as they are strongly influenced by features such as non-stationary and asymmetric behaviour, non-linearity, and high persistence of conditional variance (Zhu & Zhou, 2009). However, with the advancement of high-end computational facilities aided by vast information availability, we witness a paradigm shift in application of prediction methods. Focus has shifted to machine learning (ML)-based approaches, because of proven efficiency in deciphering hidden patterns presented in time series data. The biggest advantage is that while these methods do not essentially require any justification for distributional assumptions, they also end up reducing variance largely by addressing data biasness. Recently, Nguyen, Sermpinis, and Stasinakis (2023) depict a framework for the application of artificial intelligence and machine learning techniques for financial services, discussing the drawbacks of the symbiotic relation of data science towards financial implications.

We observe the prevalence of technical indicators in most of the studies involving stock returns (Cavalcante, Brasileiro, Souza, Nobrega, & Oliveira, 2016; Mohapatra & Misra, 2020). The observation is in line with the argument that any information corresponding to every new macroeconomic event, political news, or related phenomena is inherently reflected in stock prices. Technical analysts strongly feel that existing technical indicators are sufficient to unravel the mystery behind stock prediction problems (Nazário, e Silva, Sobreiro, & Kimura, 2017). However, another class of researchers and practitioners intensely advocate not ignoring the role of fundamental and macroeconomic indicators as they are extremely crucial in determining related impact owing to rapidly changing business cycles (Dinh, Goutte, Nguyen, & Walther, 2022; Neely, Rapach, Tu, & Zhou 2014).

Although the superiority of ML methods in predicting stock returns, vis-à-vis traditional statistical models, has already been established by existing literature (Kim, Ku, Chang, & Song, 2020), recent findings have also acknowledged the fact that a single prediction model might have its own limitations in handling bias-variance trade-off and in turn making highly precise and accurate predictions. This encourages researchers to take up ensemble ML approach, combining bagging, and boosting techniques, while addressing challenge of working with a single estimator (Sagi & Rokach, 2018). Bagging incorporates many weak learners to combine and outperform a single strong learner. It reduces variance and helps in eliminating over fitting of models. Similarly, boosting algorithms also combine several weak learners in a sequential method and help reduce high bias. In case of events, when bagging might not result in better bias, boosting will generate a model with low errors as it focuses on optimising advantages and reducing challenges in a single model. In the scenario of overfitting, bagging will be good enough to handle that.

Taking motivation from the aforesaid discussion, in the current scope of research, we consider a holistic dataset comprising technical, fundamental, as well as macroeconomic factors, influencing returns of Indian banking stocks. Recent popularity and high prediction accuracy of ensemble methods have inspired us to consider Random Forest, Gradient Boosting, and XGBoost models for predictions given the fact that these methods have proven their merit while dealing with extraordinarily complex, non-linear, non-stationary, and high-noise nature of financial data (Chong, Han, & Park, 2017; Prasad & Bakshi, 2022; Weng, Ahmed, & Megahed, 2017; Yun, Yoon, & Won, 2021).

In our study, first we train and test three models using technical factors only. We follow it up by including both technical, fundamental, and macroeconomic factors used in prediction to prove enhancement in accuracy due to the inclusion of techno-fundamental analysis. Furthermore, the study considers 20 of the largest Indian banks functional across the private and public sectors, which constitute 95% of the total market capitalisation of the Indian banking sector. To increase the robustness of our study, we do not limit our analysis to the exploration of 18 technical factors but expand the discussion to 6 fundamentals as well as 6 macroeconomic factors. Altogether, the analysis considers 30 variables.

Our current work contributes to the ML-based stock-prediction literature by combining the short-term and long-term effects of technical, fundamental, and macroeconomic variables to train and test both bagging and boosting algorithms such as Random Forest, Gradient Boosting, and XGBoost. Our paper substantiates a very high accuracy of stock-returns prediction advantages gained by the simultaneous deployment of technical, fundamental, and macroeconomic factors while establishing the predictive prowess of the ensemble ML techniques for a highly volatile stock market.

Contrary to the conventional strategy relied by the short-term positional traders, our work emphasises the role played by macro-fundamental factors such as inflation, index of industrial production (IIP), bond yield, and term spread in improving the stock price prediction accuracy, particularly during shorter time horizons. Similarly, for long-term investors, we highlight the importance of technical indicators such as moving averages (MAs) (in addition to conventional macro-fundamental variables) for timing the stock price cycles and predicting long-term returns with a higher degree of accuracy.

The results document that while technical indicators, that is price, volume, and momentum strategies, remained extremely relevant for short-term holding period returns (20 days, 40 days, and 60 days), feature importance of fundamental and macroeconomic variables was highlighted for the medium-term (180 days) and long-term (240 days) periods. We further find that both macroeconomic variables and technical indicators described varying degrees of

information while forecasting the risk premium of individual stocks. In line with this observation, we recommend a combined techno-fundamental analysis for a superior prediction, as compared to an exclusive analysis based on technical indicators or fundamental and macroeconomic variables. Based on such findings, we further emphasise that deploying ML methods radically bridges the gap between the traditional asset pricing models, depending on macroeconomic and fundamental variables, and more contemporary technical indicators. The approach adopted in this study can be tested across asset classes over different time horizons, not only to validate the effectiveness, but also the stability of the models in predicting stock returns, using technical, fundamental, and macroeconomic variables.

The rest of the paper is organised as follows. The second section presents the literature review; the third section gives the data description. The fourth section discusses the methodology and metrics considered in the present study; the fifth section discusses the empirical results and findings, and finally, in the sixth section, we conclude and provide the policy implications and suggestions for future research.

Literature review

The first decade of the century witnessed several researchers implementing neural networks and support vector machines (SVM) for stock prediction (Henrique, Sobreiro, & Kimura, 2019). O'Connor and Madden (2006) observed that neural network based trading decisions led to a return of 23.5% as compared to 13.03% return from the benchmark DJIA index between December 2002 and December 2004. Cao and Tay (2001) applied SVM techniques to the S&P 500 (between 1993 and 1995) and found it to be a better tool for forecasting S&P 500 than neural network. Huang, Nakamori, and Wang (2005) applied SVM techniques to predict the weekly direction of NIKKEI 225 and compared with the discriminant analysis techniques.

Yuan, Yuan, Jiang, and Ain (2020) applied feature selection techniques based on Random Forest, SVM, neural network to predict market direction of China between 2010 and 2018. The Random Forest model has demonstrated better performance than traditional ML models (SVM and neural network). Sadorsky (2022) used a tree-based approach to forecast the direction of stock prices in the solar sector between 2008 and 2021. Trading signals generated from Extra Trees stock price direction prediction performed better as compared to plain vanilla buy and hold strategy. Kim et al. (2020) applied Random Forest to stock price (technical) data of various MSCI indexes between January 2000 and December 31, 2018. They found an accuracy ratio in the range of 50–60%. Recently, Han, Kim, and Enke (2023) applied XGBoost to the 92 listed stocks at NASDAQ. They only considered technical stock price data and observed trading Min-Max labelling using XGBoost performed better than other prominent labelling methods. Song (2018) applied XGBoost technique to the 20 well-known stocks listed in DJIA (between 2010 and 2017) to predict future stock price direction by using historical prices and technical indicators and obtained an accuracy of 60%. Similarly, Basak, Kar, Saha, Khaidem, and Dey (2019) applied Random Forest and XGBoost to predict stock return for a horizon of 3, 5, 10, 15, 30, 60, and 90 days. They took 10 stocks listed in the U.S. stock exchange on daily data from 2000 to 2014 and found that Random Forest performed better than XGboost. They noticed that prediction accuracy increased with larger holding periods. In the case of XGBoost, it ranged from 53% (for a three-day holding period) to 77% (for a 90-day holding period). In the case of Random Forest, it ranged from around 65% (for a three-day holding period) to around 90% (for a 90-day holding period) with only technical indicators.

Selvamuthu, Kumar, and Mishra (2019) used neural networks based on three learning algorithms, namely, Bayesian Regularisation, Levenberg-Marquardt, and Scaled Conjugate Gradient for stock market prediction of Reliance Industries Limited based on tick data and 15-min data between November 30, 2017 and January 11, 2018. All the three algorithms gave an accuracy of 99.9% when the tick data was used. However, the accuracy over 15-min dataset declined to 98.9%, 96.2%, and 97.0% for Bayesian Regularisation, Levenberg-Marquardt, and Scaled Conjugate Gradient, respectively.

To cover these visible shortcomings in existing stock prediction literature, we consider a combination of technical, fundamental, and macroeconomic variables to train and test both Bagging and Boosting algorithms such as Random Forest, Gradient Boosting, and XGBoost. As the above literature suggests, these methods have performed quite well in different stock markets. In proposed framework, we have considered technical, fundamental, and macroeconomic variables to perform stock prediction related to the banking sector in an emerging market such as India, where banking sector constitutes 30% of entire market-cap of the benchmark NIFTY-50 index and is a lead indicator of the country's economic development.

Recent developments in the domain of ML based stock returns forecasting are tabulated below for a quick reference:

S. No.	Author (publication year) and journal	Dataset	Output predicted	Forecasting time period and frequency	Methodology proposed	Outcome
1	Liu and Long (2020) *Digital Signal and Processing*	S&P 500, DJIA and CMSB	Stock price prediction	Period: 17 December 2010 to 17 January 2013 Frequency: daily	Empirical wavelet transform, outlier robust extreme learning machine (ORELM) model-based post-processing, LSTM-based deep neural network (DNN)	The hybrid model has the best prediction accuracy as compared to traditional ML techniques such as long short term memory (LSTM) and Random Forest

(*continued*)

(*Continued*)

S. No.	Author (publication year) and journal	Dataset	Output predicted	Forecasting time period and frequency	Methodology proposed	Outcome
2	Shen and Shafiq (2020) *Journal of Big Data*	Chinese Stock Market (SSE and CSI-300)	Stock price trend (up/down)	Period: 1 January 1989 to 31 December 1998 Frequency: daily	LSTM + feature expansion + recursive feature elimination + principal component analysis	The proposed model performs better than techniques such as multilayer perceptron (MLP), SVM, and Random Forest
3	Lu, Li, Li, Sun, & Wang (2020) *Complexity* *IEEE Access*	Shanghai Composite index	Stock price	Period: 1 July 1991 to 31 August 2020 Frequency: daily	Convolutional neural network (CNN) + LSTM	The model performed better than individual CNN, MLP, recurrent neural network (RNN), CNN-RNN, and LSTM models
4	Carta, Consoli, Piras, Podda, and Recupero (2021) *IEEE Access*	S&P 500 index	Stock price movement	Period: 2005 to 2018 Frequency: daily	Decision tree, lexicon generation and feature extraction	The model performed better than MLP, Gradient Boosting, and Random Forest
5	Kyriakou, Mousavi, Nielsen, and Scholz (2021) *Annals of Operations Research*	S&P 500 index, consumer price index, sovereign bond yield, and long-term and short-term interest rates	Long-term stock index returns in excess of interest rate (long- and short terms) and the inflation. P/E ratio	Period: 1872 to 2015	Non-parametric smoother containing the covariates	The combination of long-term and short-term price earnings, long-term- and short-term spreads is the best-performing set of variables for predicting the annual stock returns net of inflation
6	Yıldırım, Toroslu, and Fiore (2021) *Financial Innovation*	Technical Indicator: MACD, MAs, ROC, RSI, and Bollinger Bands Fundamental variables, CPI, and interest rates	EURO/USD	Period: January 2013 to January 2018	Deep learning based LSTM model	LSTM is found to be very effective in many time-series forecasting problems and to make direction predictions in Forex
7	Wang, Tang, Xiong, and Zhang (2021) *Expert Systems with Applications*	S&P 500 index, DJI, NASDAQ and FTSE	Stock Price Movement	Period: 1 July 1991 to 31 August 2020 Frequency: daily	Reservoir computing model	The model performed better than RNN and LSTM
8	Sadorsky (2022) *The North American Journal of Economics and Finance*	Invesco Solar ETF (TAN) where TAN tracks the MAC Global Solar Energy index	Stock prices of solar index	Period: 15 April 2008 to 25 June 2021 Frequency: daily	Tree-based approach to forecast the direction of stock prices in the solar sector	Trading signals which are generated from Extra Trees stock price direction prediction have successfully performed better as compared to plain vanilla buy and hold strategy
9	Toochaei and Moeini (2023) *Expert Systems with Applications*	47 Financial ratios of stocks listed on Tehran Stock Exchange (TSE). 10 macroeconomic variables	TSE	Period: 21 March 2005 to 20 March 2020	14 different bagging- and boosting-based ensembles	1. Boosting techniques: XGBoost performed best. At the same time, Ada Boost performed worst 2. Among the bagging-based techniques, Rotation Forest performed best. At the same time, Random Forest performed worst
10	Wang, Ma, Bouri, and Guo (2023) *Journal of Forecasting*	18 technical indicators along with 17 famous macroeconomic variables, such as the realised volatility of S&P 500 and world economic activity index	Bitcoin volatility	Period: December 2011 and April 2021	Shrinkage methods, such as least absolute shrinkage and selection operator (LASSO) and elastic net	Shrinkage techniques can successfully extract predictive information from technical indicators and fundamental variables 2. The results show strong evidence that the macroeconomic indicators have a better ability to forecast Bitcoin volatility in comparison with technical indicators

Data

In this section, we classify raw data depending on their inherent nature and their contribution towards stock prediction. The three types of classification followed in the present study are as follows:

- Market data (involving technical indicators): Market data constitute technical indicators such as opening and closing prices, intra-day high and intra-day low prices, trade volume, and momentum indicators used as input parameters for explaining prediction targets and resultant trading activities.
- Fundamental data: This essentially comprises quarterly reported accounting data informing market participants about asset-liability composition, balance sheet information, and the profit and loss statement of companies, alongside firm-specific information, such as earnings per share, dividend yield, and price-to-book ratio.
- Macroeconomic data: Variables such as gross domestic product (GDP), consumer price index, IIP, and money supply, amongst others, reflect global- and country-level economic circumstances affecting the overall performance of stock market, individual sectors, and stocks in turn.

We realise that for a successful prediction, practitioners must target feeding algorithms with data deriving future returns. The feed data essentially consider factors related to future economic outlook (i.e., the fundamental factors) and the resulting demand and supply comprising technical factors. The objective is to maximise signal-to-noise ratio during the data structuring process. Hence, in the current study, we include similar factors spread across fundamental, technical, and macroeconomic realms as we try to explore their relevance in deploying the ensemble ML models for Indian banking stocks.

The study sample comprises 20 public and private sector banks[1] continuously traded on the National Stock Exchange of India between the period January 2013 and June 2022. The sample encompasses most liquid and well-capitalised banking stocks whose performance works as a lead indicator for the overall economy. As of June 2022, the total market capitalisation of banking stocks stands at a valuation of $3.5 billion. These banks constitute 94% of the market capitalisation of the entire banking sector.

For our analysis, we consider 6 macroeconomic variables, 6 fundamental variables, and 18 technical variables, in accordance with previous literature (Neely et al., 2014; Wen, Liu, Wang, & Zhang, 2022; Yin & Yang 2016). The benefit of such classification is that this set of variables helps to capture market-specific, as well as firm-specific information, along with macroeconomic events. Furthermore, 18 technical variables (which are derived from price and volume data) are based on the three major trend strategies, which are further classified as: price-based strategy, momentum-based strategy, and volume-based strategy.

[1] They are namely Axis Bank, Bank of Baroda, Bank of India, Bank of Maharashtra, Central Bank of India, City Union Bank, Federal Bank, HDFC Bank, ICICI Bank, IDBI Bank, Indian Bank, Indian Overseas Bank, IndusInd Bank, Karur Vysya Bank, Kotak Bank, Punjab National Bank, SBI bank, UCO Bank, Union Bank, and Yes Bank.

Price and volume-based technical strategy

Moving averages (MAs)

The moving averages rule based strategy generates either a sell or a buy signal, ($S_{i,t} = 1$ and $S_{i,t} = 0$) at end of time t (in days), by comparing two MAs as follows:

$$S_{i,t} = 1; \text{if } MA_{s,t} > MA_{l,t} \tag{1a}$$

$$S_{i,t} = 0; \text{if } MA_{s,t} < MA_{l,t} \tag{1b}$$

where

$$map_{j,t} = \left(\frac{1}{j}\right)\sum_{i=0}^{j-1} P_{t-1}; j = s, l \tag{2}$$

P_t is the current stock price; s is the short-term time period referring to 20, 40, and 60 days; l is the long-term time period referring to 180 and 240 days; $map_{j,t}$ is an indicator denoting MAs with lengths s and l by MA(s, l). The moving averages rule observes changes in stock price trends because short-term MAs have higher sensitivity towards recent movements in prices than that of long-term MAs. Based on the above discussion, we have developed indicators reported in Table 1.

Momentum-based strategy

These momentum-based indicators generate either a buy or a sell signal ($S_{i,t} = 1$ and $S_{i,t} = 0$, respectively) as follows:

$$S_{i,t} = 1; \text{if } P_t \geq P_{t-m} \tag{3a}$$

$$S_{i,t} = 0; \text{if } P_t < P_{t-m} \tag{3b}$$

Intuitively, if the current stock price is higher than its level m days before, then it indicates a 'positive' momentum, along with higher expected returns. Hence, it generates a buy signal. We have denoted the momentum indicator by MOM(m), which compares P_t to P_{t-m}. We have computed this signal for m = 20, 40, 60, 120, 180, and 240 days, as described in Table 2.

Volume-based strategy

Technical analysts commonly apply volume data in combination with past stock prices to study market trends. As a result, in our third strategy, we have incorporated 'on-balance volume'. First, we define:

$$OBV_t = \sum_{k=1}^{t} Vol_k D_k \tag{4}$$

where Vol_k measures trading volume during period k; D_k is the binary variable which will take the value of 1 if $P_k > P_{k-1}$ otherwise 0.

We then compute a trading signal with the help of OBV_t such as

$$S = 1; \text{if } MA_{s,t}^{OBV} > MA_{l,t}^{OBV} \tag{5a}$$

$$S = 0; \text{if } MA_{s,t}^{OBV} < MA_{l,t}^{OBV} \tag{5b}$$

Table 1 Description of price-based indicators.

S. No.	Indicator	Description
1	map(1,9)	Equals 0, if moving average stock price of previous 20 days is more than moving average stock price of previous 180 days, otherwise 1
2	map(1,12)	Equals 0, if moving average stock price of previous 20 days is more than moving average stock price of previous 240 days, otherwise 1
3	map(2,9)	Equals 0, if moving average stock price of previous 40 days is more than moving average stock price of previous 180 days, otherwise 1
4	map(2,12)	Equals 0, if moving average stock price of previous 40 days is more than moving average stock price of previous 240 days, otherwise 1
5	map(3,9)	Equals 0, if moving average stock price of previous 60 days is more than moving average stock price of previous 180 days, otherwise 1
6	map(3,12)	Equals 0, if moving average stock price of previous 60 days is more than moving average stock price of previous 240 days, otherwise 1

Note: Numbers mentioned in the parentheses represent months. In one month, we have 20 trading days.

Table 2 Description of momentum-based indicators.

S. No.	Indicator	Description
1	Mom1	Equals 0, when current stock price is lesser than previous 20-day closing price, otherwise 1
2	Mom2	Equals 0, when current stock price is lesser than previous 40-day closing price, otherwise 1
3	Mom3	Equals 0, when current stock price is lesser than previous 60-day closing price, otherwise 1
4	Mom4	Equals 0, when current stock price is lesser than previous 120-day closing price, otherwise 1
5	Mom5	Equals 0, when current stock price is lesser than previous 180-day closing price, otherwise 1
6	Mom6	Equals 0, when current stock price is lesser than previous 240-day closing price, otherwise 1

Note: Numbers mentioned in the parentheses represent months. In one month, we have 20 trading days.

where

$$MV_{j,t} = \left(\frac{1}{j}\right)\sum_{i=0}^{j-1} OBV_{t-1}; j = s, l \quad (6)$$

Instinctively, higher volumes coupled with price rises indicate a strong underlying trend generating a buy signal. We calculate the signals for s = 20, 40, and 60 days and l = 180 and 240 days as described in Table 3.

The price-based moving average indicators, momentum-based indicators, and volume-based indicators are considered as trend-following parameters (Neely et al., 2014; Sullivan, Timmermann, & White, 2001). We further discuss the fundamental and macroeconomic indicators, and the output features in the following sections.

Fundamental indicators

Fundamental indicators such as business performance, debt levels, cash flows, dividend distribution, sector outlook, and earnings management describe the intrinsic value of a firm. Fundamental indicators are relatively stable as compared to technical indicators, in response to any changes in liquidity

Table 3 Description of volume based indicator.

S. No.	Indicator	Description
1	MAV(1,9)	Equals 0, if moving average on-balance volume of previous 20 days is more than moving average on-balance volume of previous 180 days, otherwise 1
2	MAV(1,12)	Equals 0, if moving average on-balance volume of previous 20 days is more than moving average on-balance volume of previous 240 days, otherwise 1
3	MAV(2,9)	Equals 0, if moving average on-balance volume of previous 40 days is more than moving average on-balance volume of previous 180 days, otherwise 1
4	MAV(2,12)	Equals 0, if moving average on-balance volume of previous 40 days is more than moving average on-balance volume of previous 240 days, otherwise 1
5	MAV(3,9)	Equals 0, if moving average on-balance volume of previous 60 days is more than moving average on-balance volume of previous 180 days, otherwise 1
6	MAV(3,12)	Equals 0, if moving average on-balance volume of previous 60 days is more than moving average on-balance volume of previous 240 days, otherwise 1

Note: Numbers mentioned in the parentheses represent months. In one month, we have 20 trading days.

Table 4 Description of fundamental indicators.

S. No.	Indicator	Description
1	Dividend yield	Dividend disbursed as a percentage of individual share price of a stock. It is calculated by taking a log of 12 months moving sum of dividends minus log of lagged stock prices.
2	Earnings price ratio	It is calculated by taking log of 12-month moving sum of earnings on the benchmark index minus log of benchmark index prices. This ratio compares earnings of an index with respect to the price. In other words, this ratio indicates how much an investor market is giving valuation to an index/company as compared to its earnings.
3	Dividend payout ratio	Percentage of profit is disbursed as dividend. It is calculated by taking log of 12-month moving sum of dividends of benchmark index minus log of a 12-month moving sum of benchmark index earnings.
4	Equity risk premium	It measures how much excess return index is providing over a risk-free investment. Total return index of benchmark index minus 10-year Government of India bond yield. Total return index refers to total return given by a stock or index which includes capital appreciation as well as dividend distributed.
5	Price to book ratio	Price to book ratio for benchmark index. Book values refer to net asset value of company. Price refers to market price of a company. This ratio indicates how much company is valued in a market as compared to the net value of assets a company possess.
6	Dividend price ratio	Log (benchmark index dividend yield) minus log (total return index of benchmark).

Table 5 Description of macroeconomic indicators.

S. No.	Indicator	Description
1	T-bill rate	Interest rate on a three-month Indian treasury bill. It is a short-term borrowing instrument issued by RBI on behalf of Government of India.
2	Long-term bond yield	Ten-year Government of India bond yield. It is a long-term bond which is issued by the government of India with a maturity period of 10 years.
3	Term spread	Ten-year long-term Government of India bond minus T-bill. It is an indicator of recession. If the interest rate on long-term bonds becomes smaller than interest rate on short-term bonds, then it becomes a leading indicator for recession. Ideally, a long-term bond should have higher interest rates than short-term bonds due to liquidity preference theory.
4	Inflation	Consumer price index of India. It is a monthly indicator which captures the retail side inflation of India. It is an aggregate figure which also captures energy and food prices in addition to other consumables.
5	Index of industrial production; IIP	It is a monthly data which indicates the monthly output in the manufacturing sector.
6	Money supply	It is broad money supply in Indian economy. Preciously we have considered M2 money supply which is sum of currency in circulation in the economy in addition to short-term bank deposits. It is a measure of liquidity in the economy.

or news. These indicators are mainly used by medium- and long-term investors having longer horizons (more than one year). Table 4 describes the fundamental indicators considered in our analysis.

Macroeconomic indicators

Macroeconomic variables indicate the broad economic condition and are strongly dependent on the policy formulation by local governments, central banks, and the global economy indicating the state of the business cycle, which in turn states the direction of the financial market. In the case of dynamic asset pricing, expected time-varying returns depend upon the future economic outlook. These variables track the dynamic macroeconomic conditions and thus play a critical role in predicting future stock returns (Cochrane, 1991; Neely et al., 2014). These variables are described in Table 5.

Output feature

We calculate the excess return of 20 individual banking stocks over the benchmark index return. Equation (7) calculates stock price return of 20 individual banking stock over a period of short- (20, 40, and 60 days), medium- (180 days), and long-term holding period of 240 days. Equation (8) shows stock price return of the benchmark index with a holding period of 20, 40, 60, 180, and 240 days. Equation (9) denotes the excess return of 20 bank stocks over the benchmark index for a holding period of 20, 40, 60, 180, and 240 days:

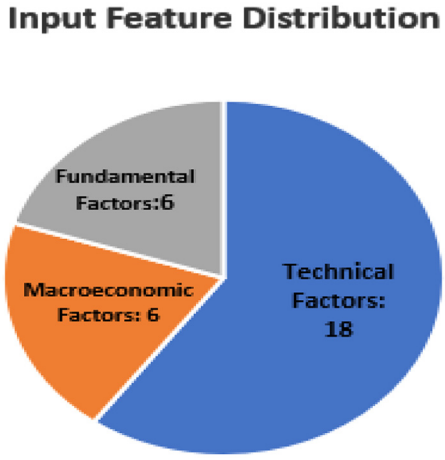


Figure 1 Input feature distribution.

$$R^s_{t.m} = \frac{P^s_{t+m}}{P^s_t} - 1 \tag{7}$$

$$R^l_{t.m} = \frac{P^l_{t+m}}{P^l_t} - 1 \tag{8}$$

$$ER^s_{t.m} = R^s_{t.m} - R^l_{t.m} \tag{9}$$

Following Equation (9), the final output is coded as

$$ERet^s_{t.m} = 0; \text{if } ER^s_{t.m} < 0 \tag{10a}$$

$$ERet^s_{t.m} = 1; \text{if } ER^s_{t.m} \geq 0 \tag{10b}$$

Prediction workflow, model design, and performance metrics

Explaining prediction workflow

Assorted raw data types have varying degrees of effort, associated with their extraction and processing. For ML models, a humongous dataset is essential for training purposes. In the current study, we strongly feel that for a superior performance of the models in consideration, a combination of market data, fundamental data, alongside with macroeconomic data, should be the appropriate choice as the combination provides the largest possible amount of sample data. The input feature distribution is shown in Figure 1.

Figure 2 portrays the design framework for the current work. The data are collected from the Eikon database provided by Thomson Reuters. The features are constructed using technical, fundamental, and macroeconomic parameters. The constructed dataset is split into a training and a testing set. Three ML methods, that is, Random Forest, XGBoost, and Gradient Boosting are implemented, and all the implemented models are trained and validated using different metrics explained in the following sub-sections.

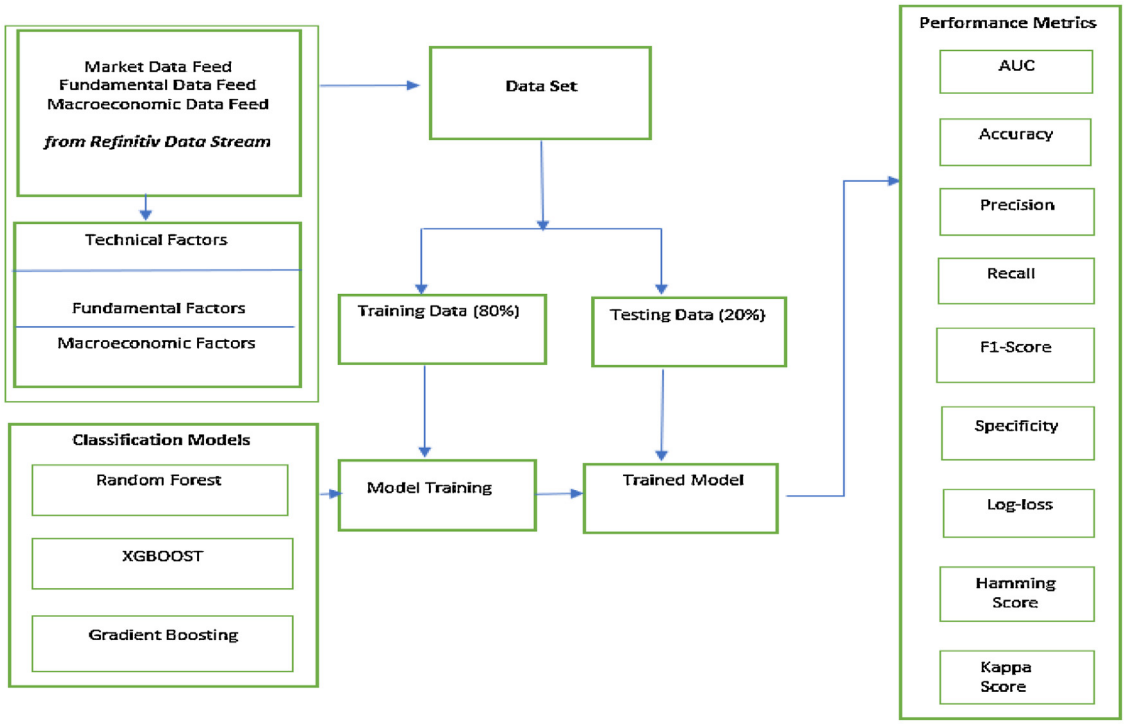


Figure 2 Explaining the prediction workflow.

Explaining the machine learning models

This sub-section explains the ML approaches and their related features for being deployed in the study.

The present stock prediction deals with a binary classification problem on tabular data. To enhance the performance of this binary problem and have a better interpretability (related to feature importance), bagging (Random Forest), and boosting algorithms (Gradient Boosting and XGBoost) based on the concepts of computer science and statistical physics are being used (Bühlmann & Hothorn, 2007).

Grinsztajn, Oyallon, and Varoquaux, (2022) have also established the superiority of tree-based models such as Random Forest, Gradient Boosting, and XGBoost over deep networks on tabular data across a large number of datasets from various domains and hyperparameter combinations.

Random Forest can address the non-linear dynamics in the data. The prediction prowess of the model is based on the ensemble technique where the forest combines many decision trees while performing as a cohort. The simple philosophy is the wisdom of the crowd where the trees save each other from individual error, resulting in the minimum at the end. While some trees may be classified wrongly, others may be correct, adding up the entire thing in the right way.

Boosting is a technique that combines multiple weak classifiers into a single strong learning model by optimising the error function. The Boosting algorithm introduces a facility to optimise a cost function that can be defined by the user and is almost specific to the problem (Zhong & Enke, 2019). Further, prediction accuracy is achieved in XGBoost algorithm (an advanced version of Gradient Boosting) in many cases as compared to Gradient Boosting through a series of properties such as regularisation, high flexibility to generate custom optimisation objectives, and evaluation criteria (Ozcan, Sakar, & Suloglu, 2020). More about the three models considered in the present work has been elaborated in the following section.

Random Forest

Random Forest is an ML algorithm that is applied for classification problems. It comprises a few hundred decision trees; those are constructed taking observations from the dataset. It is ensured that there is no correlation among the trees, and hence, there will be less chance of overfitting. Each tree is generally a sequence of a Yes/No question, depending on one or a number of variables. At each node of a tree (corresponding to a single question), the data are divided into two sets, where similar observations are grouped in one category. Hence, the importance of each feature or variable is generated from the purity of each of the divided sets (Breiman, 2001; Zhong & Enke, 2019).

Gradient Boosting

In Gradient Boosting algorithm, the output of multiple weak learners, typically decision trees is combined sequentially. The new models are trained to predict errors generated by previous models. The Gradient Boosting model comprises a loss function which is required to be optimised, an underperformed learner to predict, and a module to add outputs from under-performed learners to minimise the errors. The weights are updated by computing the negative gradient of the loss function with respect to the predicted output (Friedman, Hastie, & Tibshirani, 2002).

XGBoost

XGBoost (Extreme Gradient Boosting) is an improvised version of the Gradient Boosting model. In this model, a function is approximated by optimising certain loss functions and applying several regularisation methods. XGBoost can develop many models that can be combined sequentially through a gradient descent approach which contributes in terms of computation and processing resources. The model has properties of handling missing values, pruning trees, inclusive of cross-validation (Chen & Guestrin, 2016).

Explaining the performance metrics

This sub-section explains metrics used in examining the performance of the ML methods. In the present work, popular metrics, such as Precision, F1-Score, Recall, Accuracy, Hamming Score, Specificity, Kappa Score, Area under Receiver Operating Curve (AUC), and Log Loss, have been used. Since the current work involves a two-class classification problem, the outcomes of the classification model can be presented by statistical metrics, such as True Negative (TN), True Positive (TP), False Negative (FN), and False Positive (FP). These four metrics constitute the confusion matrix.

- TP represents the output whenever the model predicts the positive class correctly.
- TN represents the output whenever the model predicts the negative class correctly.
- FP represents the outcome whenever the actual class is negative, but the model predicts the opposite.
- FN represents the outcome whenever the actual class is positive, but the model predicts the opposite.

The parameters such as Recall, Precision, and F1-Score are computed as follows:

- *Recall:* This metric represents how correctly the TPs are identified by the model. It is calculated as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}$$

- *Precision:* This metric represents the proportion of positive predictions that are actually correct. It can be computed as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}$$

- *F1-Score:* It is the harmonic mean of Precision and Recall given by:

$$\text{F1} - \text{Score} = \frac{2\times(\text{Recall}\times\text{Precision})}{\text{Recall}+\text{Precision}}$$

- *Accuracy:* This metrics represents the fraction of prediction done correctly by the model and is measured as:

Table 6 Comparison of ML models for 20-day holding period returns using technical indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log loss	Hamming Score	Kappa Score
XGBoost	0.77	0.69	0.69	0.69	0.69	70	0.58	0.31	0.37
Random Forest	0.76	0.68	0.68	0.68	0.68	0.67	0.71	0.31	0.37
Gradient Boosting	0.67	0.62	0.62	0.62	0.62	0.64	0.64	0.37	0.24

Table 7 Comparison of ML models for 40-day holding period returns using technical indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log Loss	Hamming Score	Kappa Score
XGBoost	0.86	0.77	0.77	0.77	0.77	0.78	0.46	0.22	0.54
Random Forest	0.85	0.77	0.77	0.77	0.77	0.77	0.58	0.22	0.54
Gradient Boosting	0.81	0.73	0.73	0.73	0.73	0.74	0.52	0.26	0.46

Table 8 Comparison of ML models for 20-day holding period returns using technical indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log Loss	Hamming Score	Kappa Score
XGBoost	0.92	0.85	0.85	0.85	0.85	0.85	0.35	0.15	0.69
Random Forest	0.92	0.84	0.84	0.84	0.84	0.84	0.44	0.15	0.69
Gradient Boosting	0.88	0.81	0.81	0.81	0.81	0.80	0.43	0.18	0.62

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

- *AUC:* This metric is considered as a good measure for comparison of the performance of classifiers. It is the AUC and is calculated as follows:

$$\text{AUC} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right)$$

- *Hamming Score:* This metric is computed by performing XOR between predicted and actual labels followed by averaging across the dataset. It presents the proportion of correct predictions compared to total labels.
- *Kappa Score:* This metric supports the comparative performance of the models. Comparison is based only on accuracy which may mislead with bias. Kappa Score inspects classifications that may be due to chances. Its value ranges from −1 to 1. As its value approaches 1, the performance of the model is more realistic (McHugh, 2012).

Results and findings

The experimental results are demonstrated in two sub-sections. The first sub-section compares the prediction power of ML methods, and the second sub-section represents the different feature important plots generated by the above discussed ML methods to determine the best performing models measured against the defined metrics in the above section.

Comparison of ML methods

We compare the performance of the ML methods in two different stages. In stage 1 of the comparative study, we explore the prediction accuracy of the models using only technical indicators, while in stage 2, we include the fundamental and the macroeconomic factors, alongside the technical factors to understand the improvements (if any). The study is conducted for 20-, 40-, 60-, 180-, and 240-day holding period returns.

Comparing ML methods based on technical parameters

We implement three different ML models, namely Random Forest, Gradient Boosting, and XGBoost by employing Google Colab with a Jupyter Notebook environment and the GPU runtime environment with Python v3.8.5. The data are divided into a ratio of 80:20 for training and testing purposes. Packages such as Pandas, Numpy, Matplotlib, and Sklearn have been used for implementations. GridSearchCV, a function of Sklearn's model_selection package, has been used for tuning the hyperparameters. Ten-fold cross-validation has also been used. The results from the classifiers are presented with nine performance metrics as described above. Tables 6—10 report

Table 9 Comparison of ML models for 180-day holding period returns using technical indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log Loss	Hamming Score	Kappa Score
XGBoost	0.97	0.92	0.92	0.91	0.91	0.92	0.23	0.08	0.82
Random Forest	0.96	0.91	0.91	0.91	0.91	0.92	0.33	0.08	0.82
Gradient Boosting	0.94	0.89	0.89	0.89	0.89	0.89	0.28	0.10	0.79

Table 10 Comparison of ML models for 240-day holding period returns using technical indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log Loss	Hamming Score	Kappa Score
XGBoost	0.85	0.78	0.77	0.78	0.77	0.76	0.46	0.22	0.54
Random Forest	0.85	0.77	0.77	0.77	0.77	0.76	0.55	0.22	0.55
Gradient Boosting	0.81	0.73	0.73	0.73	0.73	0.72	0.46	0.26	0.46

the comparison of the implemented ML models for certain holding periods, that is, 20, 40, 60, 180, and 240 days using only technical indicators.

From the above results, we observe underperformance of models for ultra-short-term (20-day) and short-term (40-day) holding period returns. We find the performance metrics, essentially Accuracy Score, F1-Score, Precision values, Recall score, and Specificity numbers lie between a modest range of 62%—78%. Furthermore, we learn that while the models' performance significantly improves for mid-term holding periods (i.e., more than 80% for a 60-day period and 90% for a 180-day holding period returns, respectively), there is a steep decline for longer holding period of 240 days. We conclude that by deploying solely technical indicators the models deliver their best only for mid-term holding periods of 60 and 120 days.

Taking a cue from the aforesaid discussion, it becomes imperative for us to include firm-specific fundamental variables and the macroeconomic factors, alongside the technical indicators and re-examine the prediction accuracy ratios for the ML models, particularly during ultra-short-term (20 days) and short-term periods (40 days). The related experiment and the corresponding results are discussed below.

Comparing ML methods based on technical, fundamental, and macroeconomic parameters

In this section, we compare ML models by introducing a combination of technical, fundamental, and macroeconomic variables. The results across various holding periods are tabulated below (refer Tables 11—15). If we consider two sets of results, that is, one comprising only technical indicators (refer Tables 6—10) and compare it against the second set comprising a combination of technical, fundamental, and macroeconomic variables (refer Tables 11—15), we observe that models exhibit a superior performance after inclusion of the fundamental and macroeconomic factors. We note that irrespective of the holding period involved, prediction accuracy for Random Forest and XGBoost model is over and above 90% across various performance metrics. This is a significant improvement as and when compared to the prediction accuracy values obtained by deploying solely the technical indicators. A similar performance improvement is seen for Gradient Boosting method involving a techno-fundamental analysis, although the overall accuracy is lesser as compared to the Random Forest and XGBoost methods.

XGBoost model outperforms the other two boosting and bagging models. In the case of XGBoost model, performance metrics, such as Accuracy, F1-Score, Precision, Recall, Log Loss, and Hamming Score, outperform the Random Forest metric scores. The performance of Gradient Boosting follows that of Random Forest (refer Table 12) where the holding period is 40 days. Table 13 presents the comparison of ML methods for a holding period of 60 days. The metrics of XGBoost still have values superior to other models. Gradient Boosting still falls behind Random Forest by a difference of around 10% in most of the metrics, such as AUC, Accuracy, F1-Score, Precision, Recall, and Specificity.

For long holding periods shown in Tables 14 and 15, the observation related to the performance remains broadly like short and medium terms. XGBoost still outperforms other models. Random Forest scores second, followed by Gradient Boosting. The finding is consistent with the findings of Akyildirim, Nguyen, Sensoy, and Šikić (2021), who established the supremacy of XGBoost in predicting returns for long-term analysis.

In the current study, the superiority of XGBoost is strongly established in both the context irrespective of whether the

Table 11 Comparison of ML models for 20-day holding period returns using technical, fundamental, and macroeconomic indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log lLoss	Hamming Score	Kappa Score
XGBoost	0.97	0.91	0.91	0.91	0.91	0.91	0.22	0.08	0.83
Random Forest	0.96	0.90	0.90	0.90	0.90	0.90	0.29	0.09	0.81
Gradient Boosting	0.88	0.79	0.79	0.79	0.79	0.80	0.45	0.20	0.59

Table 12 Comparison of ML models for 40-day holding period returns using technical, fundamental, and macroeconomic indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log Loss	Hamming Score	Kappa Score
XGBoost	0.98	0.94	0.94	0.94	0.94	0.94	0.15	0.05	0.88
Random Forest	0.98	0.93	0.93	0.93	0.93	0.94	0.21	0.06	0.86
Gradient Boosting	0.90	0.82	0.82	0.82	0.82	0.83	0.39	0.17	0.64

Table 13 Comparison of ML models for 60-day holding period returns using technical, fundamental, and macroeconomic indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log Loss	Hamming Score	Kappa Score
XGBoost	0.98	0.93	0.93	0.93	0.93	0.93	0.18	0.06	0.86
Random Forest	0.97	0.92	0.92	0.92	0.92	0.91	0.27	0.07	0.84
Gradient Boosting	0.90	0.82	0.82	0.82	0.82	0.83	0.39	0.17	0.65

Table 14 Comparison of ML models for 180-day holding period returns using technical, fundamental, and macroeconomic indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log Loss	Hamming Score	Kappa Score
XGBoost	0.97	0.91	0.91	0.91	0.91	0.91	0.22	0.08	0.83
Random Forest	0.96	0.90	0.90	0.90	0.90	0.90	0.29	0.09	0.81
Gradient Boosting	0.91	0.83	0.83	0.84	0.83	0.87	0.37	0.16	0.67

Table 15 Comparison of ML models for 240-day holding period returns using technical, fundamental, and macroeconomic indicators.

	AUC	Accuracy	F1-Score	Precision	Recall	Specificity	Log Loss	Hamming Score	Kappa Score
XGBoost	0.97	0.91	0.91	0.91	0.91	0.91	0.24	0.08	0.82
Random Forest	0.96	0.90	0.90	0.90	0.90	0.91	0.31	0.09	0.80
Gradient Boosting	0.91	0.83	0.83	0.83	0.83	0.87	0.38	0.16	0.66

analysis involves only the technical indicators or a combination of technical, fundamental, and macroeconomic factors. If one checks for shorter holding periods (20 and 40 days), accuracy and other metrics such as F1-Score and Recall values increase from a range of around 69—77% to 91—94%. For a holding period of 60 days, the prediction accuracy of XGBoost increases from 85% to 92%. A change in the phenomenon is observed for a medium-term holding period of 180 days, where a similar accuracy is observed and no major gains in prediction metrics were achieved by the inclusion of fundamental and macroeconomic variables. But again, for a longer holding period of 240 days, the accuracy metrics change from 78% to 91% for the XGBoost model. Similar enhancement in the performance of Random Forest is observed by inclusion of fundamental and macroeconomic variables.

As per the results presented in this sub-section, we observe XGBoost to be the best performing in predicting stock returns of Indian Banks, irrespective of the time periods. The accuracy of XGBoost can be contributed to its ability of boosting weak learners by optimised algorithms and efficient parallel computing. XGBoost is developed on the baseline of Gradient Boosting framework by algorithmic enhancement and system level optimisation. A cycle is followed in the algorithm where in the first stage, existing models are tested on the validation set, followed by further addition of models for increasing prediction accuracy. At the end, the algorithm tests new models, along with existing ones for a validation set, and the cycle continues till we arrive at an optimal value. Overall, XGBoost is outstanding in the case of structured or tabular data. The performance of Random Forest is appreciable and close to that of XGBoost. Random Forest can address the non-linear dynamics in data. Furthermore, ensembling different trees within the model makes it robust in prediction by reducing the total error generated by the Random Forest model.

Comparison of feature importance plots of ML models for different time periods

Here, we present the feature importance plots for the XGBoost algorithm, implemented for different holding periods in Figure 3. Irrespective of holding period duration, the XGBoost model performs consistently well with respect to other models as demonstrated in Tables 6—15. The plots for the XGBoost model are considered to judge the impact of technical and fundamental feature space. The benefit of applying ML methods, such as XGBoost, helps in comparing relative importance of different factors, unlike traditional asset pricing methods (Cochrane,1991; Fama & French,1993). XGBoost selects important features by repeatedly checking the dataset and interpreting the relationship by retaining the features. This algorithm shows the underlying mechanism across the input features and target outcomes based on the importance of optimal features.

As per the feature importance plot shown in Figure 3, we observe that macroeconomic factors, such as index of industrial production (IIP), inflation, money supply, treasury bill, and long-term bond yields, play an important role in predicting excess returns of Indian banks over the benchmark index during ultra-short-term and short-term horizons of 20, 40, and 60 days. The short-run price movements are used by intraday traders or swing traders. Any changes in

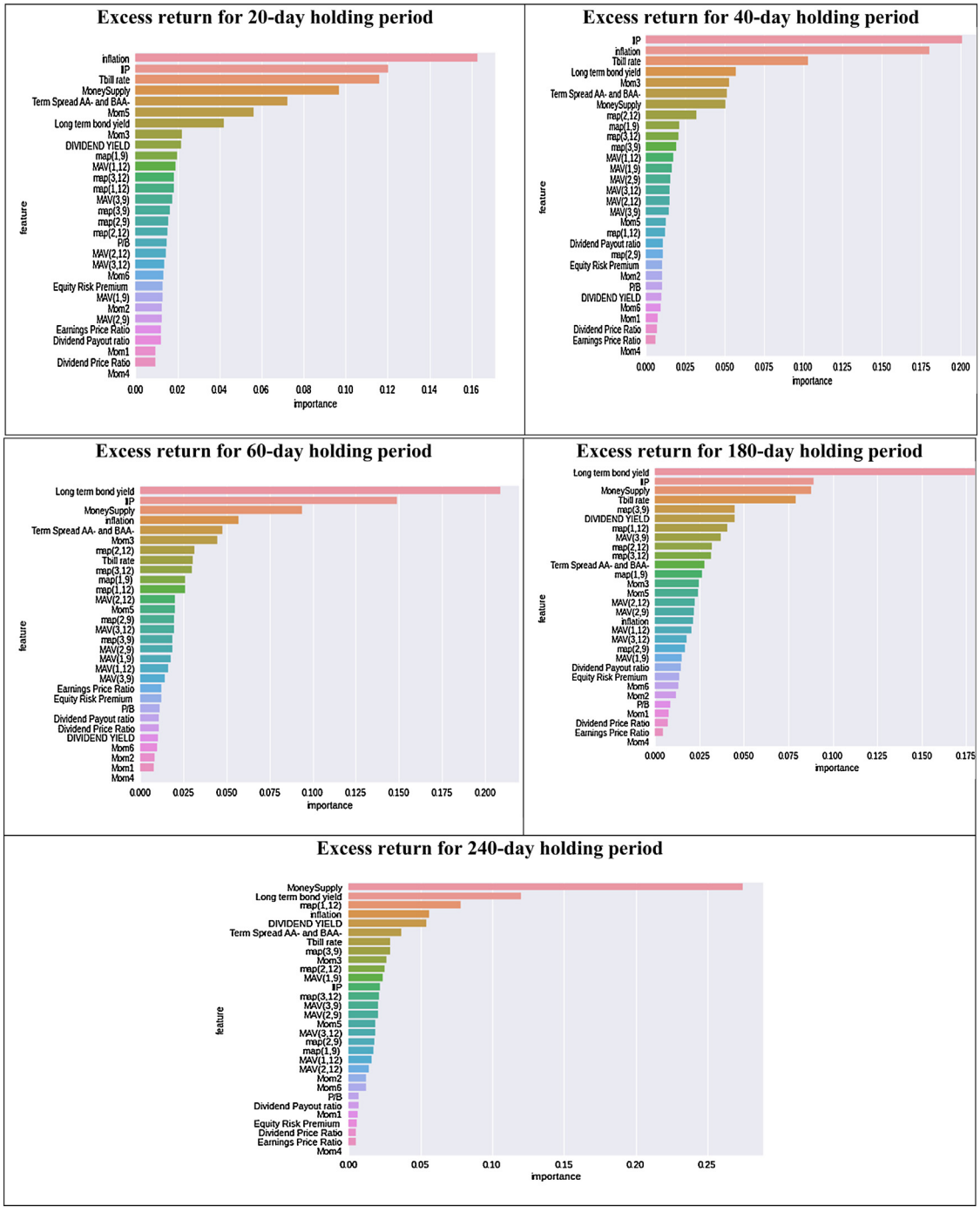


Figure 3 Feature importance plot of the different ML models for different time periods.

macroeconomic factors impact the future economic outlook and the related market expectations. As market participants are forward looking, changes in the macroeconomic variables are immediately reflected in stock prices (Long, 1974).

An increase in the feature importance of IIP indicates robust future economic growth, in turn increasing the prospects of firms registering higher income and net profits resulting in positive stock prices. Any increase in money supply by the central bank increases liquidity in the economy. Investors with increased liquidity give a higher valuation to a firm for a given level of earnings and profits, thereby positively influencing stock prices. Banking stocks are very sensitive to any change in prevailing interest rates, as interest rate movements impact their treasury income and the creditworthiness of their borrowers. Hence, banking sector stocks are impacted by movements in treasury bills and long-term bond yields. As mentioned in Table 5, term structure is a crucial recession indicator. If short-term bond yields get higher than long-term bond yields, then the yield curve suffers from an inversion adversely affecting the economic outlook and the banking sector stock performance.

In addition to the above-mentioned macroeconomic parameters, we observe that both short- (Mom3) and medium-term momentum (Mom5) are important predictors of banking stock returns for the holding periods of 20,40, 60, and 180 days. This result is in line with stock prediction literature. For long-run holding period returns (i.e., 180 and 240 days), the parameter map (2,12) and map (1,12), an indicator of the long-run price moving average impacting excess returns of banks, are found to be more important. In stock market trading, the 200-day exponential moving average is a crucial decision point for stock buying and selling for long-term investors (Ellis & Parbery, 2005; Marshall, Nguyen, & Visaltanachoti, 2017). In addition, map (3,9) plays a very important role in predicting the excess stock returns for a 180-day holding period. This finding indicates that in the long run, investors value stocks based on their fundamentals, which are typically reflected in their quarterly earnings (Chiarella, He, & Hommes, 2006; Zhu & Zhou, 2009).The results for the long-run period also establish the importance of macroeconomic factors, such as inflation, money supply, treasury bill, and long-term bond yields which reinforce the role of macroeconomic factors in long-term investing. In addition to the technical and macroeconomic factors, one noticeable observation is the role played by fundamental factors in predicting excess returns of banking stocks, for a horizon of 180 and 240 days. We observe that a stable dividend yield, not only signifies a consistency in profit sharing but also indicates good corporate governance with sound firm's fundamentals, which strengthen the long-term investors' belief (Campbell & Shiller,1988).

Discussion and managerial implications

The study has contributed to two aspects. First, it established the effectiveness of applying Bagging and Boosting techniques in predicting the stock returns in India. Second, it has a huge implication for the portfolio managers as well as the short-term traders as indicated by the feature importance plots.

In Figure 3, we observe that contrary to the traditional practice followed by the short-term investor relying heavily on the prediction power of the technical indicators, the pivotal role played by the fundamental as well as macroeconomic factors in predicting the short-term stock returns during the 20-day as well as 40-day holding periods is noteworthy. The finding can be attributed to the earnings announcements and the frequent investor meetings called by the management detailing the future course of action and related guidance. Any unexpected performance by the firms reflected in the values of the firm-specific and macroeconomic factors lead to an immediate adjustment of the stock price and the related returns owing to a knee-jerk overreaction by the short-term market participants. The overreaction can be attributed to the discounting mechanism followed by the market and any major deviation from the street expectation gets immediately incorporated into the asset prices. This phenomenon is consistent with the theory of the efficient market hypothesis. Similarly, the periodic commentary by the central banks on the inflation scenario, job market situation, and economic growth sets the undertone for the economic outlook and gives an indication of how companies and individual households will perform on economic fronts subsequently impacting the stock markets.

Further, we observe that variables such as bond yield spread, the level of the index of the industrial production impact the cost of funds and credit growth of the economy at large. Any positive surprise in the expected values of the aforesaid parameters is followed by near-term credit growth and earnings of the firms impacting their stock prices in the short term. Similarly, inflation data influence the central bank's monetary policy influencing the prime lending rates of the economy at large. Realising the 'feature importance' of the macroeconomic and fundamental factors on the firm's earnings, the short-term traders as well as the long-term investors react quickly in response to any change in the macro-fundamental information for maximising the returns on investment.

Similarly, the relative importance of the macroeconomic and the fundamental factors does not curtail the role of the technical indicators. We observe that the volume and price moving averages (MAs) alongside the momentum play a crucial role in short-term returns prediction. We find that MAs play a critical role in determining short-run 'support' and 'resistance' zone for upward- and downward-trending stocks, respectively. While co-movement of 'price' and 'momentum' in the same direction confirms trend continuation, any deviation in the direction indicates trend reversal.

For longer holding periods (60, 180, and 240 days), the macroeconomic variables are found to be having a greater impact as compared to the fundamental factors then followed by the technical indicators. This can be attributed to the fact that while fundamental investors essentially track the macroeconomic factors at large influencing the long-term earning potential of the stocks, the technical variables, such as 200-day MAs or 9-month MAs, or 50-week MAs, play a crucial role in clearly demarcating the support and resistance points. The cross-over of the price with these long-term MAs decides whether the stock is in a bullish phase or in a bearish phase helping both short-term traders as well as long-term investors.

We understand that by applying the ensemble bagging and boosting algorithms alongside the widely prevalent three types of market indicators (i.e., technical, fundamental, and macroeconomic factors), both the short-term traders as well as the long-term investors can significantly improve their prediction of the stock prices helping them to maximise their investment returns.

Conclusion

The foundation of any successful market trading strategy, involving a buy or a sell decision, is primarily based on a sound prediction of stock price movements. The complexity involved in prediction further increases as multiple factors, such as technical indicators, firm-specific fundamental parameters, and macroeconomic variables, play their crucial role. When it comes to stock market prediction, technical and fundamental analyses are poles apart. While fundamental analysis emphasises examining the intrinsic value of an underlying asset, which largely depends on historical and present information related to the firm's

revenues, cash flows and the creation of the assets and liabilities, it provides very limited help for short-term and intra-day traders who massively rely on quantitative information, such as the underlying asset price, the volume turnover, and historical returns.

Considering the explosion in the amount of data involved, with this study we attempted to predict stock price returns of the Indian banks during the period January 2013 to March 2022 using ML-based methodologies. To attain this goal, we deployed widely accepted ML methods, namely Random Forest, Gradient Boosting, and XGBoost, for measuring prediction accuracy across short-term, medium-term, and long-term holding period returns. To best exploit the predictive power of the ML methods, the analysis used 30 variables for 20 Indian banking stocks, constituting not only the technical indicators but also the macroeconomic and fundamental parameters affecting those stock returns.

At the highest level, we observe that the deployment of ML models significantly improves in explaining factors and the associated stock returns across periods. The models ingest a humongous dataset and help not only short-term traders but also long-term positional investors in solving the real-life investment problems linked to market timing and the choice of stocks.

The results documented that while technical indicators, that is, price, volume, and momentum strategies, remained extremely relevant for short-term holding period returns (20, 40, and 60 days), the feature importance of fundamental and macroeconomic variables was highlighted for the medium- (180 days) and long-term (240 days) periods. We further found that both macroeconomic variables and technical indicators described varying degrees of information, while forecasting the risk premium of individual stocks. In line with this observation, we recommend a combined techno-fundamental analysis for a superior prediction, as compared to an exclusive analysis based on technical indicators or fundamental and macroeconomic variables. Based on such findings, we further emphasise that deploying ML methods radically bridges the gap between the traditional asset pricing models, depending on macroeconomic and fundamental variables, and more contemporary technical indicators. The approach adopted in this study can be tested across asset classes over different time horizons, not only to validate the effectiveness, but also the stability of the models in predicting stock returns, using technical, fundamental, and macroeconomic variables.

Declaration of competing interest

The authors declare that they have no competing interests. Data will be provided on request. The paper has not received any funding from any organisation. All author(s) read and approved the final manuscript.

References

Akyildirim, E., Nguyen, D.K., Sensoy, A., & Šikić, M. (2021). Forecasting high-frequency excess stock returns via data analytics and machine learning. *European Financial Management* 29 (1), 22–75.

Basak, S., Kar, S., Saha, S., Khaidem, L., & Dey, S.R. (2019). Predicting the direction of stock market prices using tree-based classifiers. *The North American Journal of Economics and Finance* 47 (2019), 552–567.

Breiman, L. (2001). Random forests. *Machine Learning* 45, 5–32.

Bühlmann, P., & Hothorn, T. (2007). Boosting algorithms: Regularization, prediction and model fitting. *Statistical Science* 22 (4), 477–505.

Campbell, J.Y., & Shiller, R.J. (1988). Stock prices, earnings, and expected dividends. *The Journal of Finance* 43 (3), 661–676.

Cao, L., & Tay, F.E. (2001). Financial forecasting using support vector machines. *Neural Computing & Applications* 10 (2), 184–192.

Carta, S., Consoli, S., Piras, L., Podda, A.S., & Recupero, D.R. (2021). Event detection in finance using hierarchical clustering algorithms on news and tweets. *PeerJ Computer Science* 7, e438.

Cavalcante, R.C., Brasileiro, R.C., Souza, V.L., Nobrega, J.P., & Oliveira, A.L. (2016). Computational intelligence and financial markets: A survey and future directions. *Expert Systems With Applications* 55 (2016), 194–211.

Chen, T., & Guestrin, C. (2016, August). XG-Boost: A scalable tree boosting system. In *Proceedings of the 22nd ACM International Conference on knowledge discovery and data mining*. 785—794.

Chiarella, C., He, X.-Z., & Hommes, C. (2006). A dynamic analysis of moving average rules. *Journal of Economic Dynamics and Control* 30 (9—10), 1729–1753.

Chong, E., Han, C., & Park, F.C. (2017). Deep learning networks for stock market analysis and prediction: Methodology, data representations, and case studies. *Expert Systems with Applications* 83, 187–205.

Cochrane, J.H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance* 46 (1), 209–237.

Dinh, T., Goutte, S., Nguyen, D.K., & Walther, T. (2022). Economic drivers of volatility and correlation in precious metal markets. *Journal of Commodity Markets* 28, 100242.

Ellis, C.A., & Parbery, S.A. (2005). Is smarter better? A comparison of adaptive, and simple moving average trading strategies. *Research in International Business and Finance* 19 (3), 399–411.

Fama, E.F., & French, K.R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33 (1), 3–56.

Friedman, J.H., Hastie, T., & &Tibshirani, R. (2002). Computational statistics and data analysis. *Stochastic Gradient Boosting* 38, 367–378.

Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems* 35, 507–520.

Han, Y., Kim, J., & Enke, D. (2023). A machine learning trading system for the stock market based on N-period Min-Max labeling using XGBoost. *Expert Systems with Applications* 211, 118581.

Henrique, B.M., Sobreiro, V.A., & Kimura, H. (2019). Literature review: Machine learning techniques applied to financial market prediction. *Expert Systems with Applications* 124, 226–251.

Huang, W., Nakamori, Y., & Wang, S.Y. (2005). Forecasting stock market movement direction with support vector machine. *Computers & Operations Research* 32 (10), 2513–2522.

Kyriakou, I., Mousavi, P., Nielsen, J.P., & Scholz, M. (2021). Forecasting benchmarks of long-term stock returns via machine learning. *Annals of Operations Research* 297 (1), 221–240.

Kim, S., Ku, S., Chang, W., & Song, J.W. (2020). Predicting the direction of US stock prices using effective transfer entropy and machine learning techniques. *IEEE Access* 8, 111660–111682.

Liu, H., & Long, Z. (2020). An improved deep learning model for predicting stock market price time series. *Digital Signal Processing* 102, 102741.

Long Jr, J.B. (1974). Stock prices, inflation, and the term structure of interest rates. *Journal of Financial Economics* 1 (2), 131–170.

Lu, W., Li, J., Li, Y., Sun, A., & Wang, J. (2020). A CNN-LSTM-based model to forecast stock prices. *Complexity* 2020, 1–10.

Marshall, B.R., Nguyen, N.H., & Visaltanachoti, N. (2017). Time series momentum and moving average trading rules. *Quantitative Finance* 17 (3), 405–421.

McHugh, M.L. (2012). Interrater reliability: The kappa statistic. *Biochemia Medica* 22 (3), 276–282.

Mohapatra, S., & Misra, A.K. (2020). Momentum returns: A portfolio-based empirical study to establish evidence, factors and profitability in Indian stock market. *IIMB Management Review* 32 (1), 75–84.

Nazário, R.T.F., Silva, J.L., Sobreiro, V.A., & Kimura, H. (2017). A literature review of technical analysis on stock markets. *The Quarterly Review of Economics and Finance* 66, 115–126.

Neely, C.J., Rapach, D.E., Tu, J., & Zhou, G. (2014). Forecasting the equity risk premium: the role of technical indicators. *Management Science* 60 (7), 1772–1791.

Nguyen, D.K., Sermpinis, G., & Stasinakis, C. (2023). Big data, artificial intelligence and machine learning: A transformative symbiosis in favour of financial technology. *European Financial Management* 29 (2), 517–548.

O'Connor, N., & Madden, M. G. (2006). A neural network approach to predicting stock exchange movements using external factors. In: *Applications and innovations in intelligent systems XIII: Proceedings of AI-2005, the twenty-fifth SGAI International Conference on innovative techniques and applications of artificial intelligence*, Cambridge, UK, *2005* (pp. 64—77). London: Springer.

Ozcan, S., Sakar, C.O., & Suloglu, M. (2020). Human resources mining for examination of R&D progress and requirements. *IEEE Transactions on Engineering Management* 68 (5), 1372–1387.

Prasad, A., & Bakhshi, P. (2022). Forecasting the direction of daily changes in the India VIX index using machine learning. *Journal of Risk and Financial Management* 15 (12), 552.

Rasekhschaffe, K.C., & Jones, R.C. (2019). Machine learning for stock selection. *Financial Analysts Journal* 75 (3), 70–88.

Sadorsky, P. (2022). Forecasting solar stock prices using tree-based machine learning classification: How important are silver prices? *The North American Journal of Economics and Finance* 61, 101705.

Sagi, O., & Rokach, L. (2018). Ensemble learning: A survey. *Wiley Interdisciplinary Reviews. Data Mining and Knowledge Discovery* 8 (4), e1249.

Selvamuthu, D., Kumar, V., & Mishra, A. (2019). Indian stock market prediction using artificial neural networks on tick data. *Financial Innovation* 5 (1), 1–12.

Shen, J., & Shafiq, M.O. (2020). Short-term stock market price trend prediction using comprehensive deep learning system. *Journal of Big Data* 7 (1), 1–33.

Song, Y. (2018). Stock trend prediction: Based on machine learning methods (Doctoral dissertation, UCLA). ProQuest. https://escholarship.org/uc/item/0cp1×8th

Sullivan, R., Timmermann, A., & White, H. (2001). Dangers of data mining: The case of calendar effects in stock returns. *Journal of Econometrics* 105 (1), 249–286.

Toochaei, M.R., & Moeini, F. (2023). Evaluating the performance of ensemble classifiers in stock returns prediction using effective features. *Expert Systems with Applications* 213, 119186.

Wang, W.-J., Tang, Y., Xiong, J., & Zhang, Y.-C. (2021). Stock market index prediction based on reservoir computing models. *Expert Systems with Applications* 178, 115022.

Wang, J., Ma, F., Bouri, E., & Guo, Y. (2023). Which factors drive Bitcoin volatility: Macroeconomic, technical, or both? *Journal of Forecasting* 42 (4), 970–988.

Wen, D., Liu, L., Wang, Y., & Zhang, Y. (2022). Forecasting crude oil market returns: Enhanced moving average technical indicators. *Resources Policy* 76, 102570.

Weng, B., Ahmed, M.A., & Megahed, F.M. (2017). Stock market one-day ahead movement prediction using disparate data sources. *Expert Systems with Applications* 79, 153–163.

Yin, L., & Yang, Q. (2016). Predicting the oil prices: Do technical indicators help? *Energy Economics* 56, 338–350.

Yıldırım, D.C., Toroslu, I.H., & Fiore, U. (2021). Forecasting directional movement of Forex data using LSTM with technical and macroeconomic indicators. *Financial Innovation* 7, 1–36.

Yuan, X., Yuan, J., Jiang, T., & Ain, Q.U. (2020). Integrated long-term stock selection models based on feature selection and machine learning algorithms for China stock market. *IEEE Access* 8, 22672–22685.

Yun, K.K., Yoon, S.W., & Won, D. (2021). Prediction of stock price direction using a hybrid GA-XGBoost algorithm with a three-stage feature engineering process. *Expert Systems with Applications* 186, 115716.

Zhong, X., & Enke, D. (2019). Predicting the daily return direction of the stock market using hybrid machine learning algorithms. *Financial Innovation* 5 (1), 1–20.

Zhu, Y., & Zhou, G. (2009). Technical analysis: An asset allocation perspective on the use of moving averages. *Journal of Financial Economics* 92 (3), 519–544.